

Graph Learning with GNNs

Applications and limitations

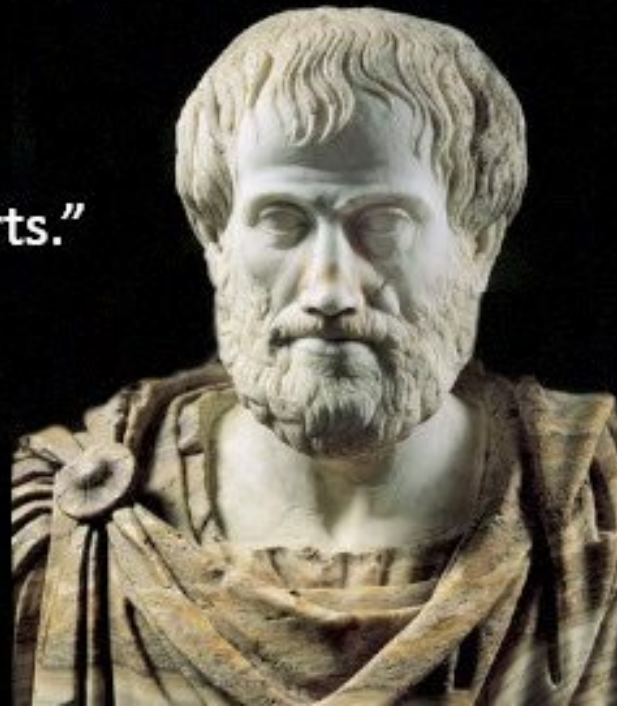
Ali Hariri

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“The whole is greater
than the sum of its parts.”

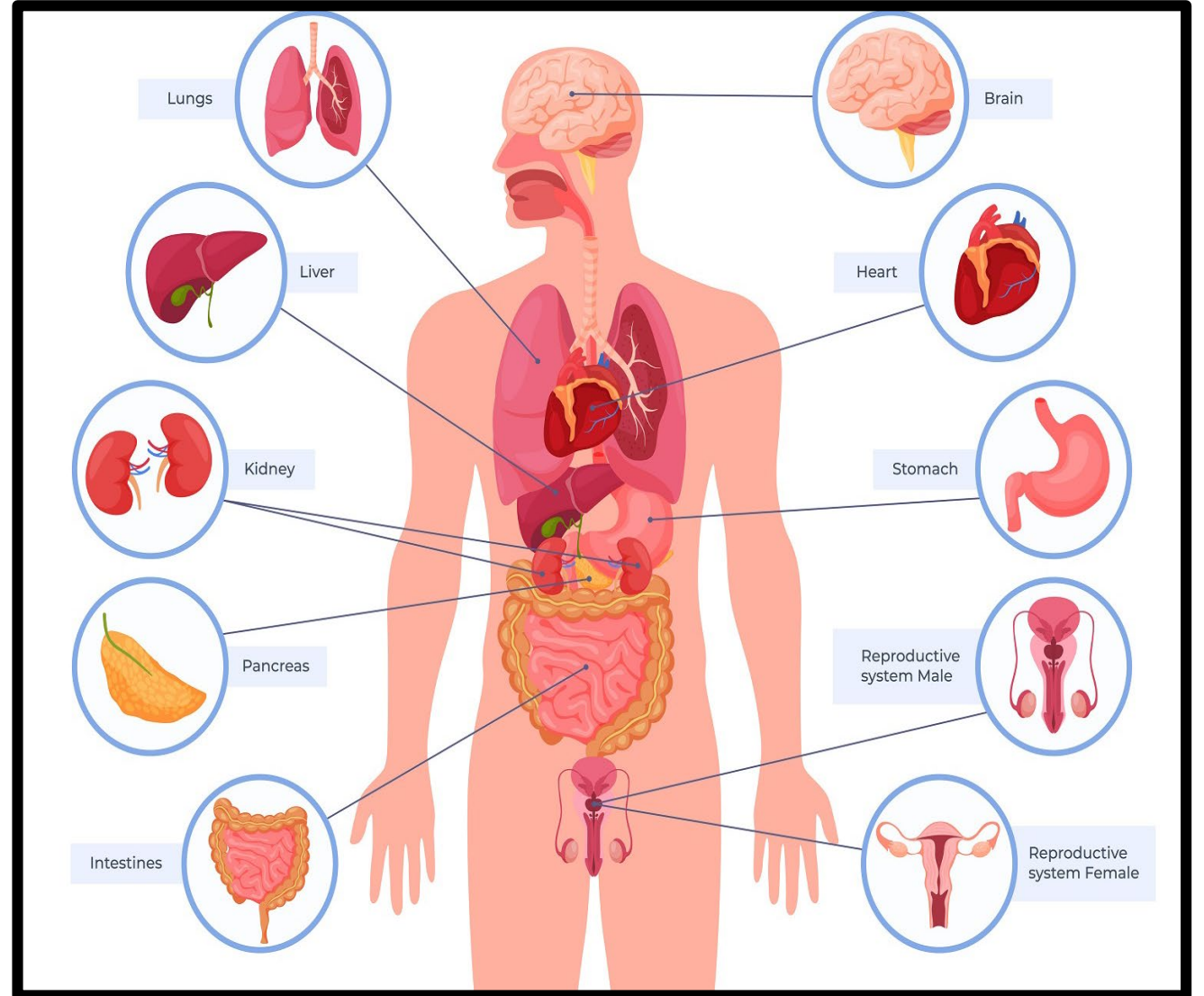
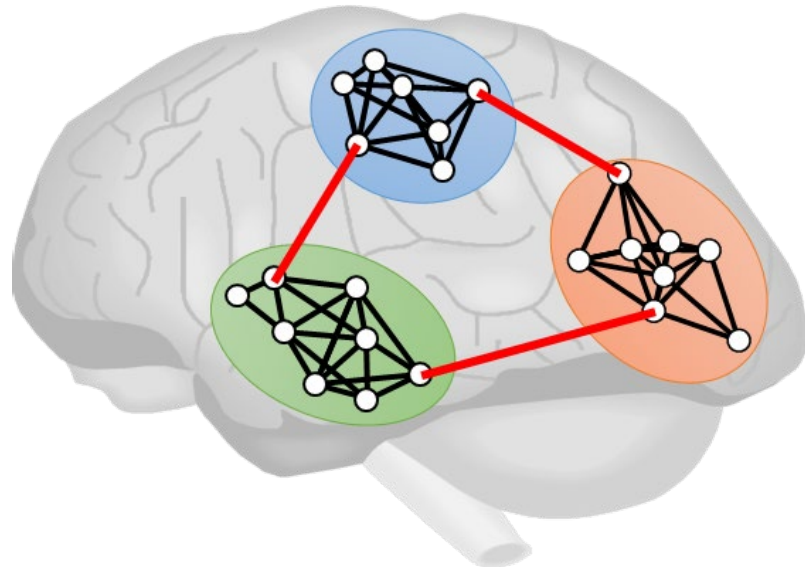
-Aristotle



Are we still talking about graphs ?

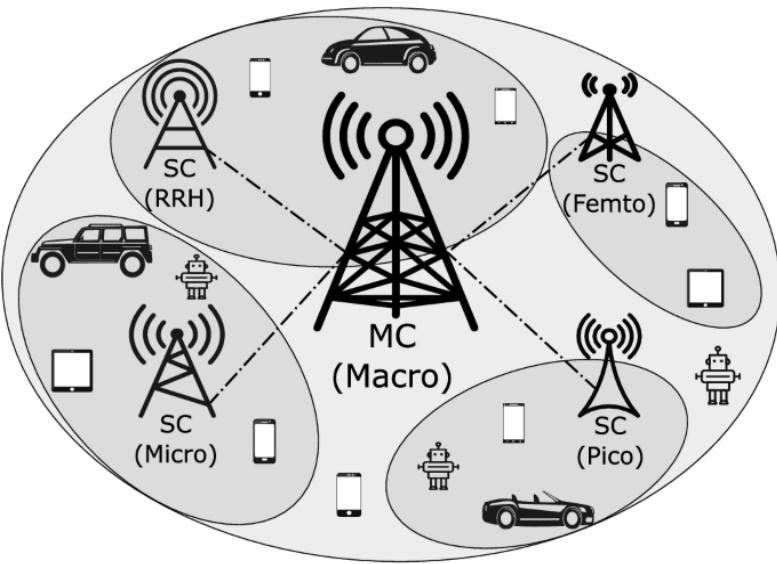


“The whole is greater than the sum of its parts”



Our world is full of different types of networks whose communication is interesting to study

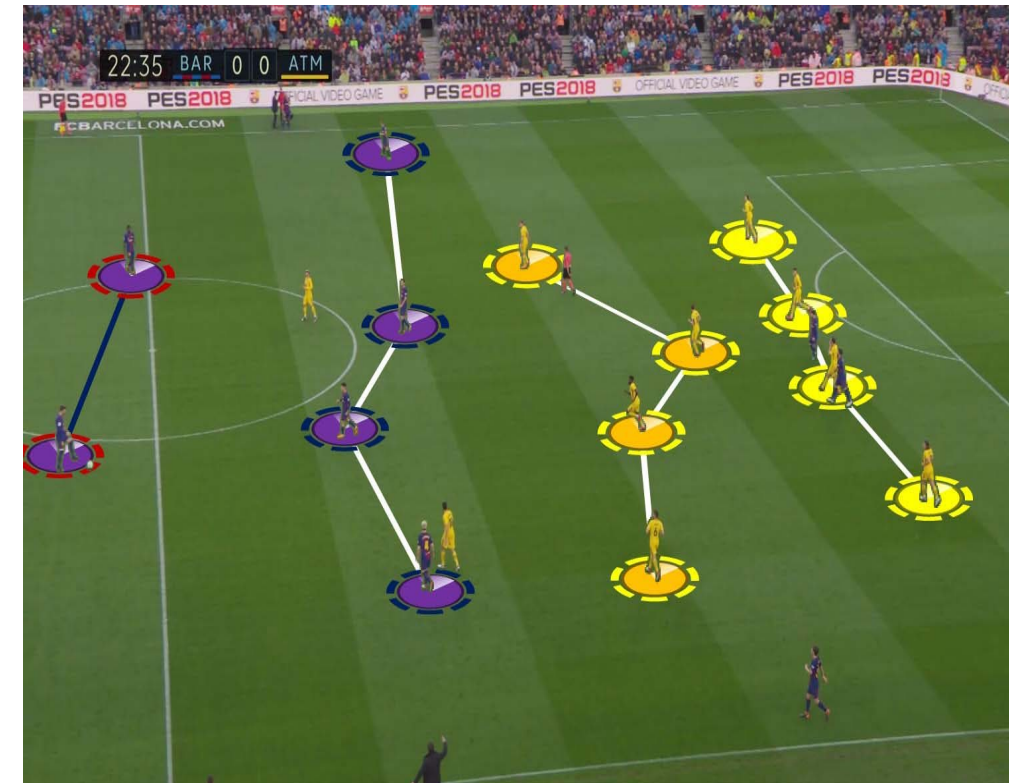
Technology Sector



Animal world

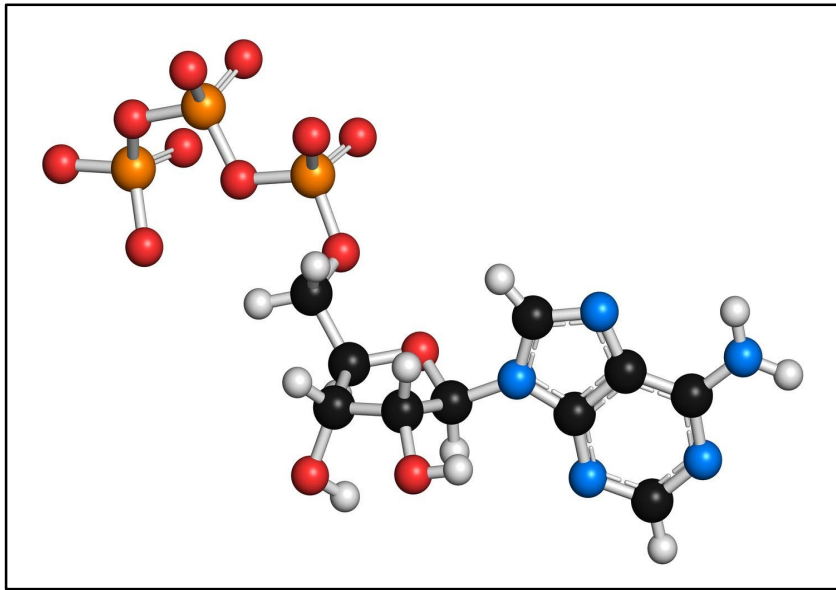


Human networks !



Graphs structures are more common than we think !

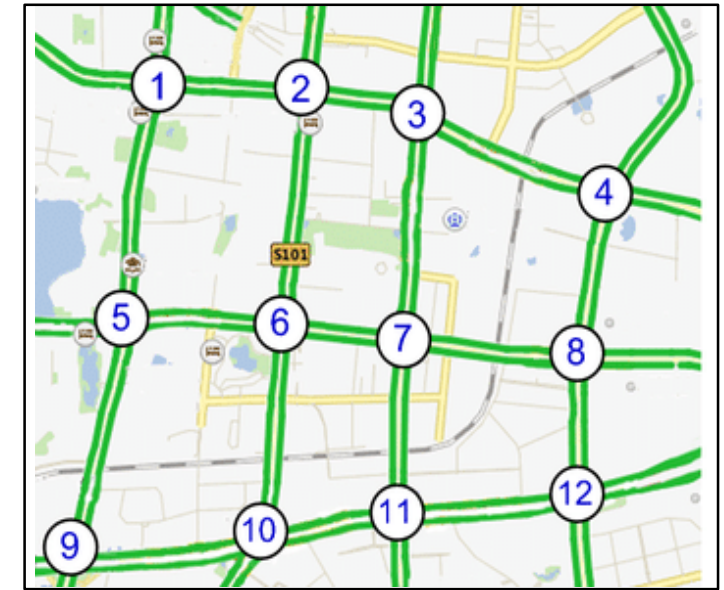
**Molecular
compounds**



**Social
Networks**

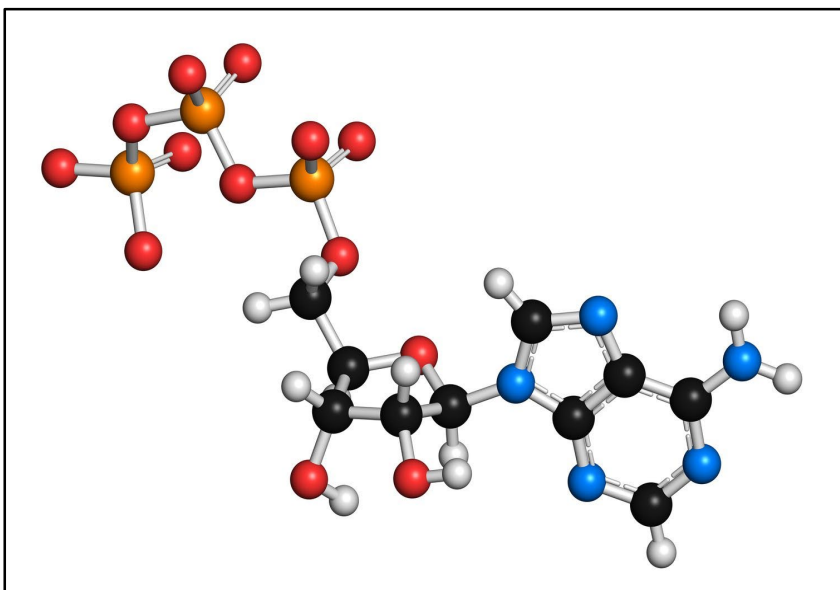


**Traffic
Routes**



We can learn different tasks on graphs

**Graph-level
classification**



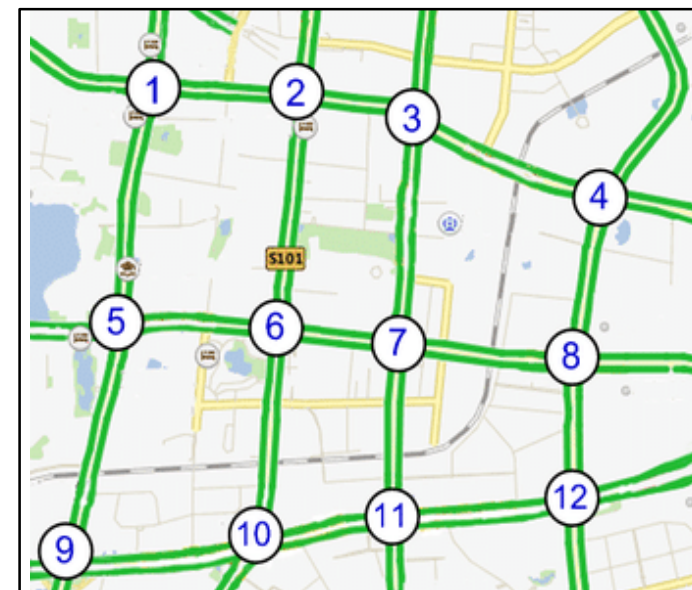
Classify the family of this compound

**Node-level
tasks**

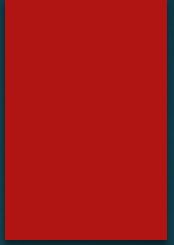


Recommend a connection for John based on common interests.

**Node-level
regression**

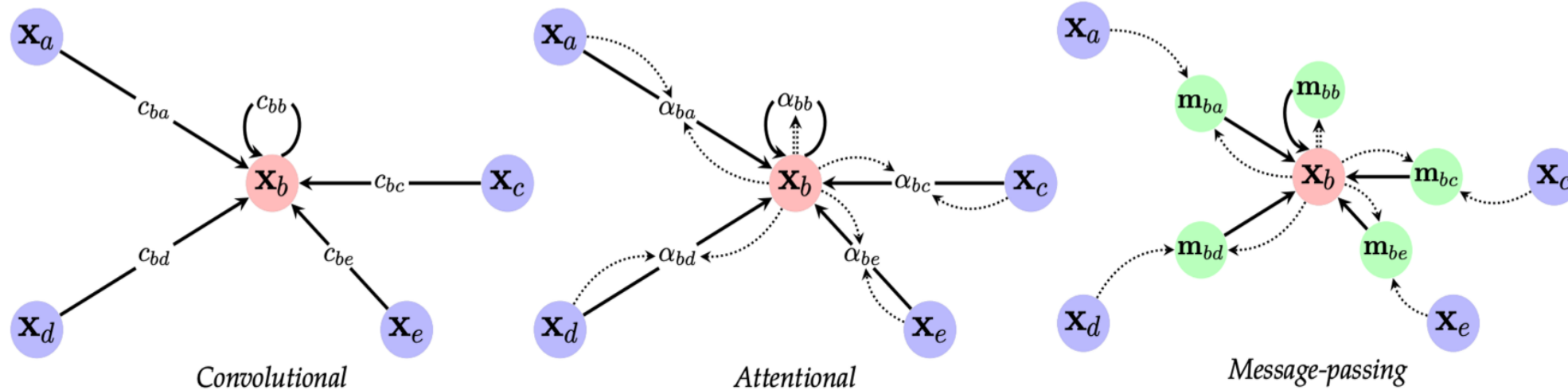


What is the average traffic at noon near Lausanne Gare ?



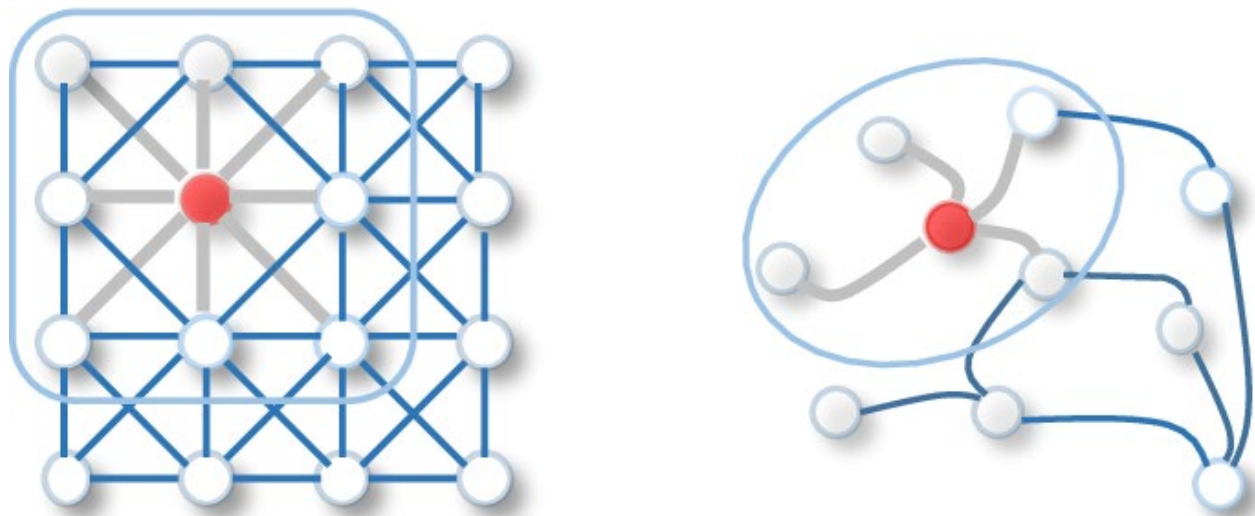
Message Passing Neural Networks (MPNNs)

Message Passing Neural Networks



- Messages are aggregated from neighboring nodes.
- Occurs over 1-hop neighborhood at a time.
- Several variations of GNNs, most of which are based on message passing.

Graph Convolutional Net (GCN)



$$H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)})$$

where:

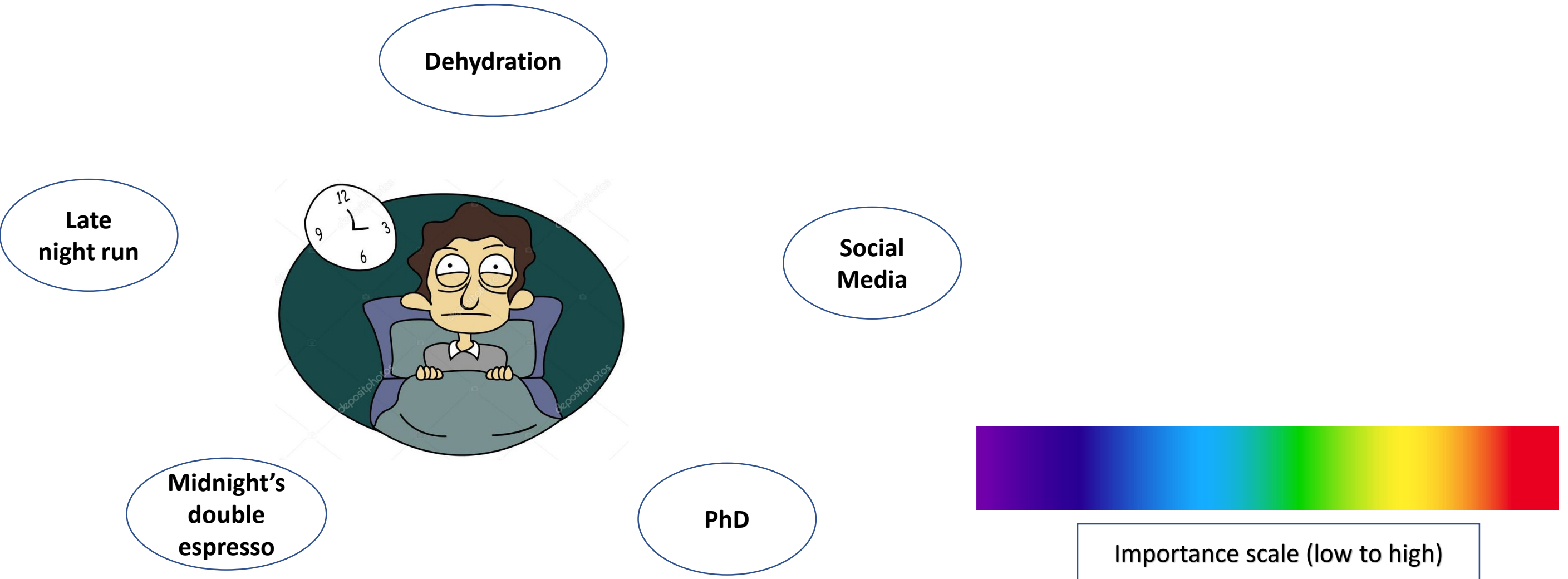
- $\tilde{A} = A + I$ (adding self-loops),
- $\tilde{D}$ is the degree matrix of $\tilde{A}$,
- $H^{(l)}$ is the node representation at layer l ,
- $W^{(l)}$ is a learnable weight matrix.



All surroundings are equal!

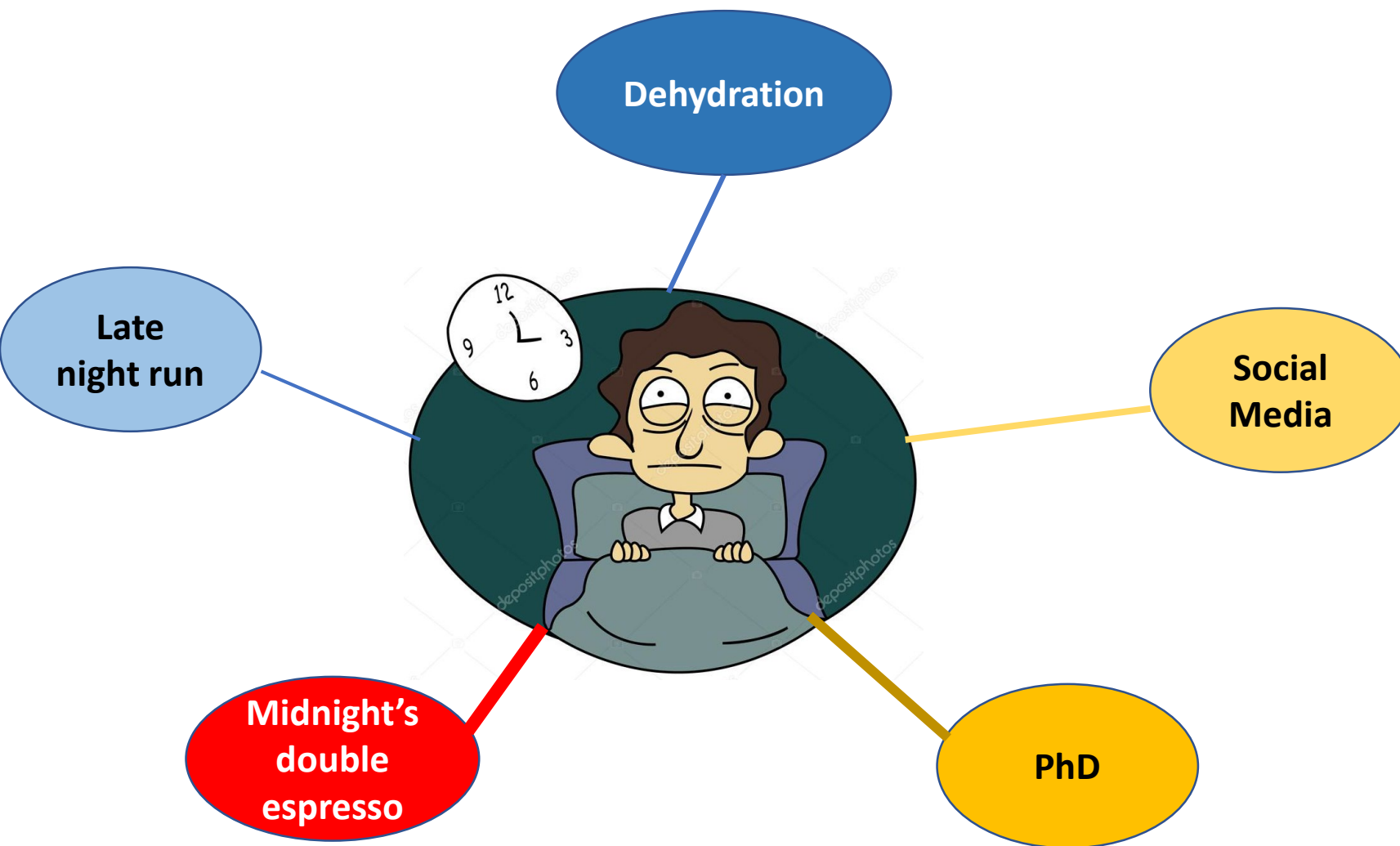
GAT – Insomnia example

Which factors contribute to Adam's bad sleep ?



GAT – Insomnia example

Let's give each factor the **attention** it requires.



GAT Formulation

$$h_v^{(t+1)} = \sigma \left(\sum_{u \in \mathcal{N}(v)} \alpha_{vu} W h_u^{(t)} \right)$$

where α_{vu} is an attention coefficient learned via a softmax function.



Importance scale (low to high)

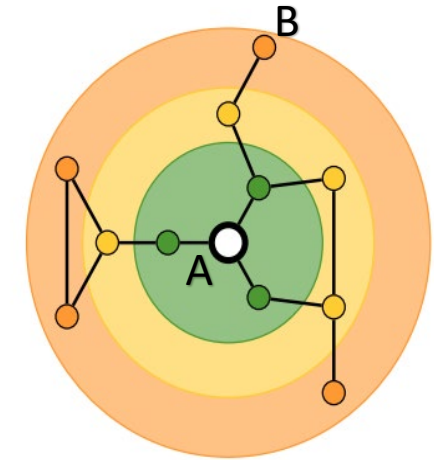
Multi-hop exploration

- Traditional GNNs aggregate information from **immediate neighbors**.
- Many real world problems require information from **further distant nodes**.

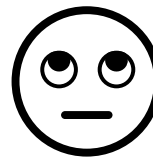
Problem: Node A is at distance K from Node B.

How does a GNNs ensure communication between them?

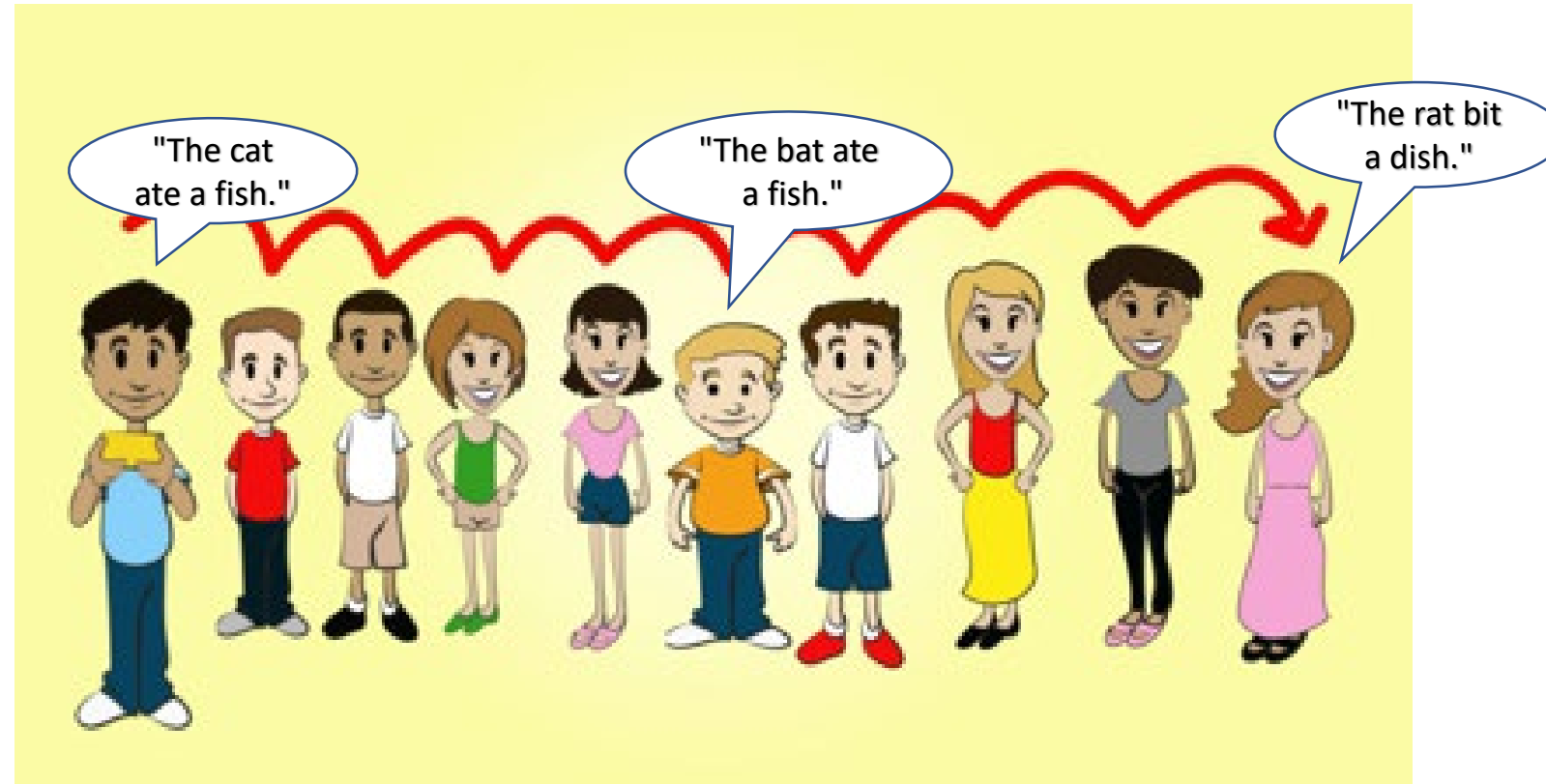
➤ **Trivial solution:** Keep adding more layers until K is reached.



What could go wrong ?

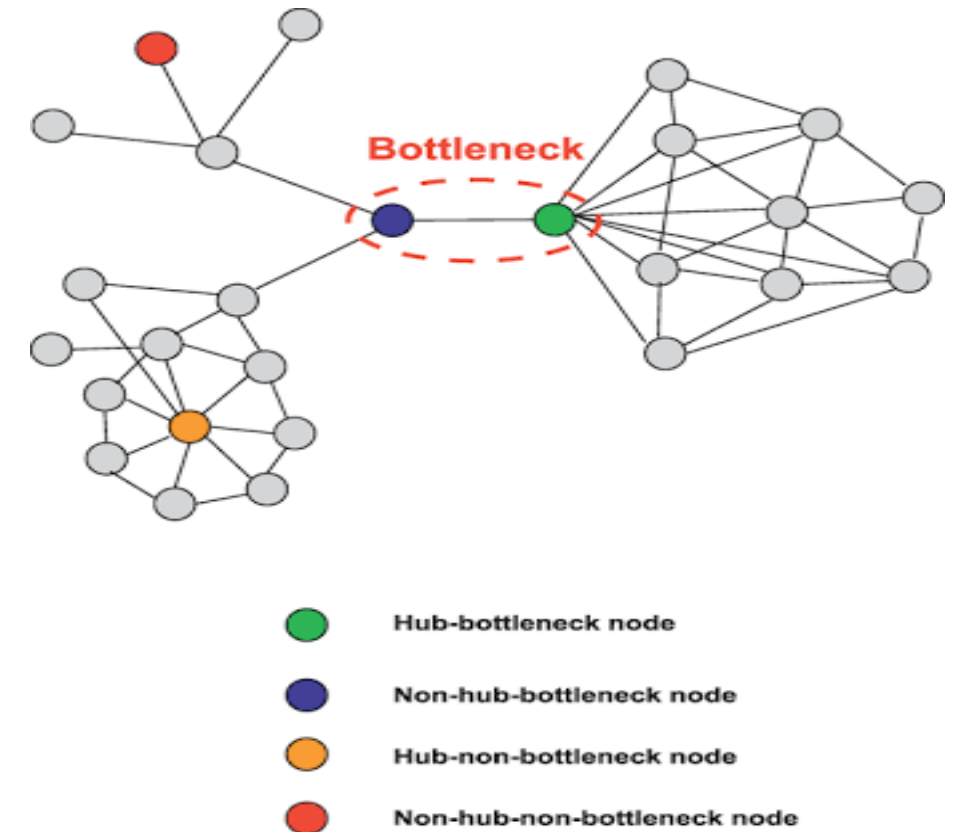


Information flow is crucial among nodes of a graph



Information flow is crucial among nodes of a graph

- Similar to transportation networks, graph neural networks are affected by traffic jams, **i.e bottlenecks**

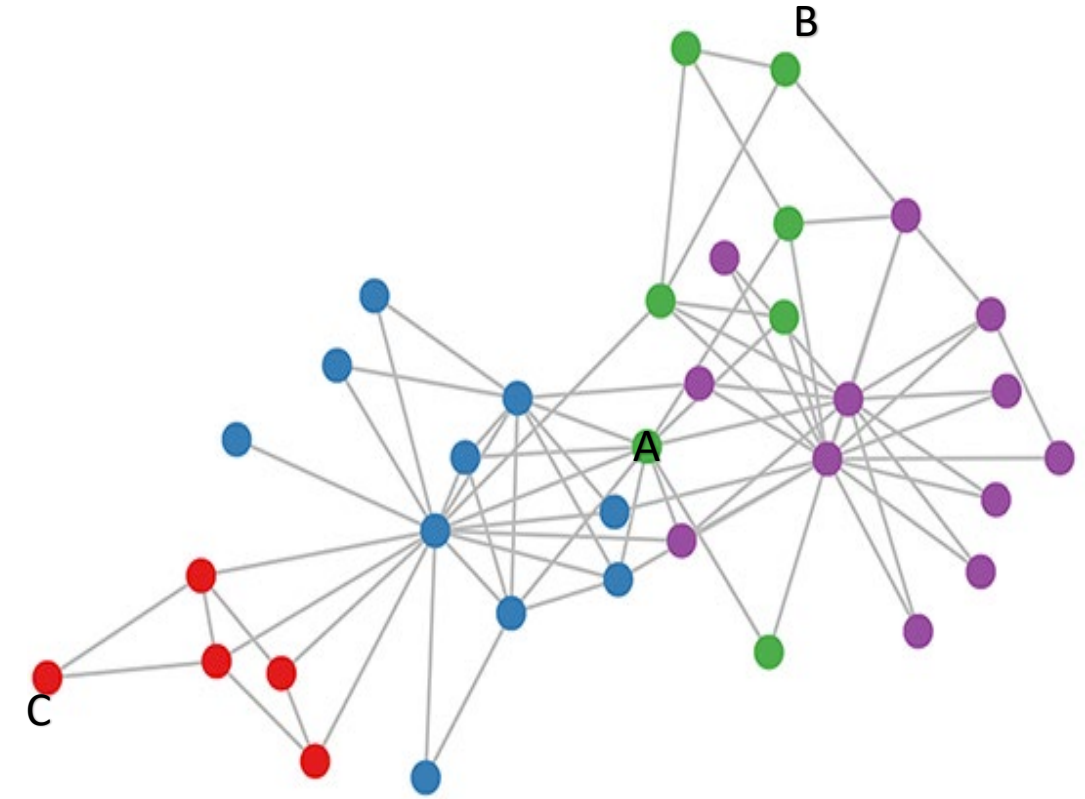




Limitations of MPNNs

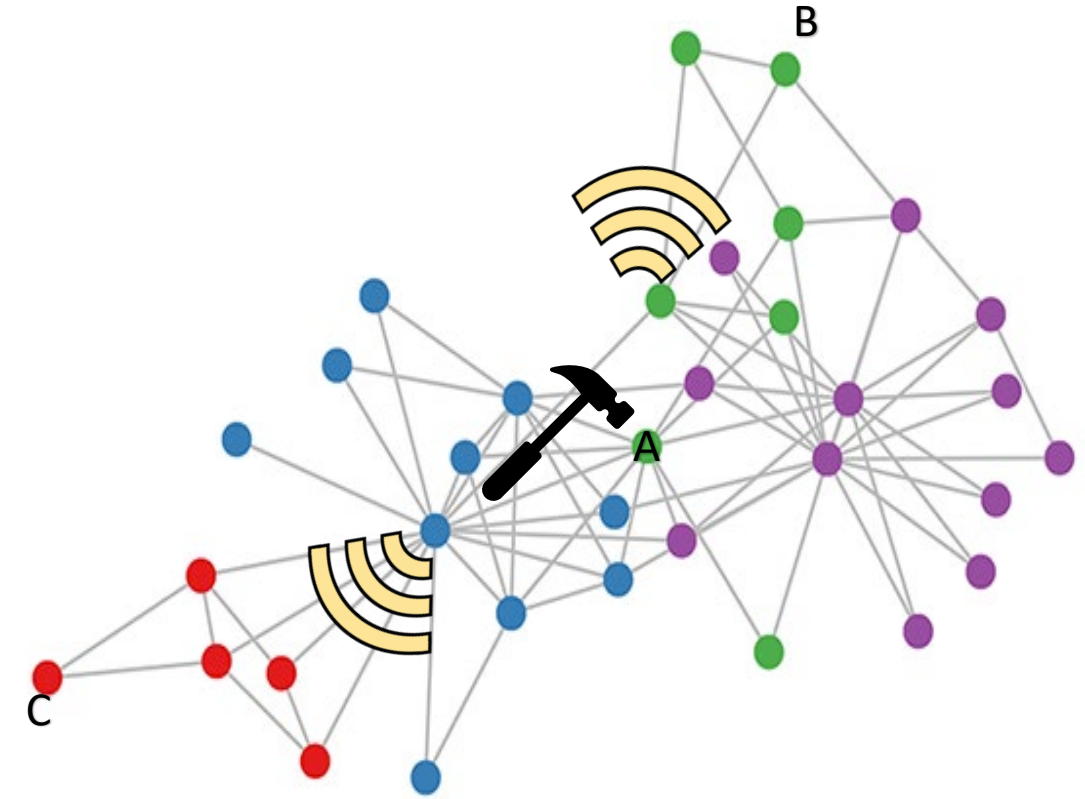
Limitations of MPNNs- Oversquashing

- ❖ Nodes can get stuck in traffic when traveling from Point A → Points B or C.
- ❖ Since many nodes are flowing to Point B through multiple routes, it is hard to distinguish **which source nodes were the most meaningful**.
- ❖ Information get “squashed” in bottlenecks while flowing from source to destination.



Limitations of MPNNs- Oversquashing

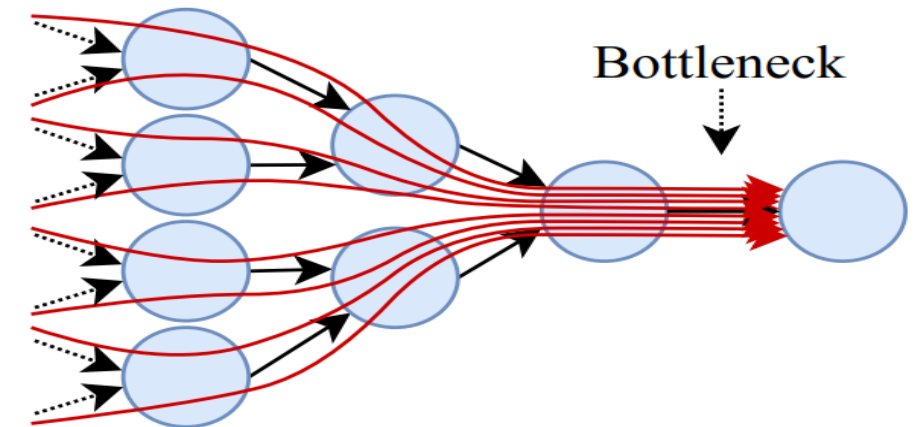
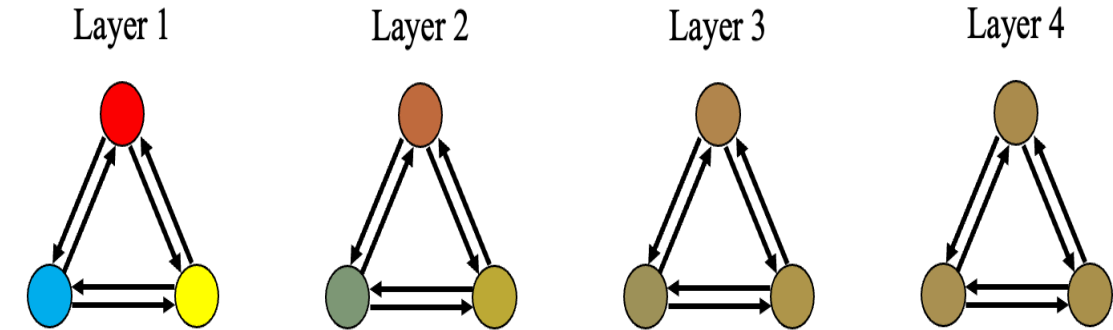
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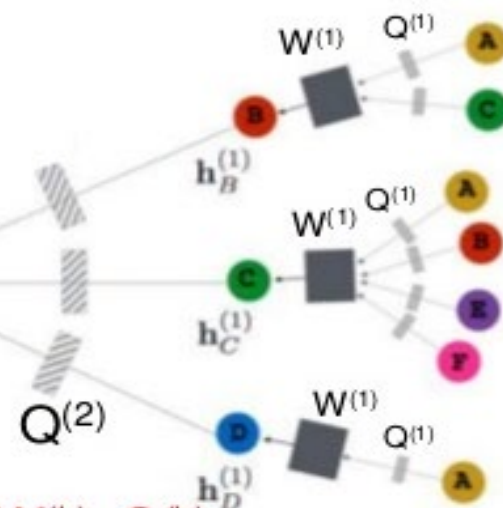
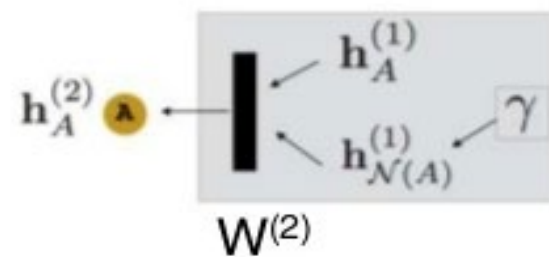
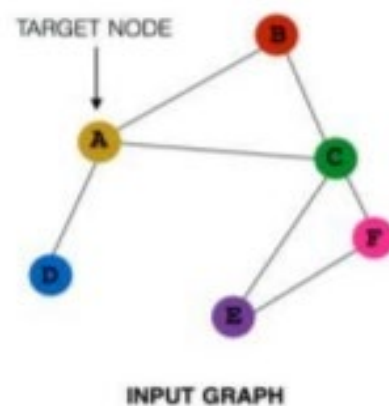
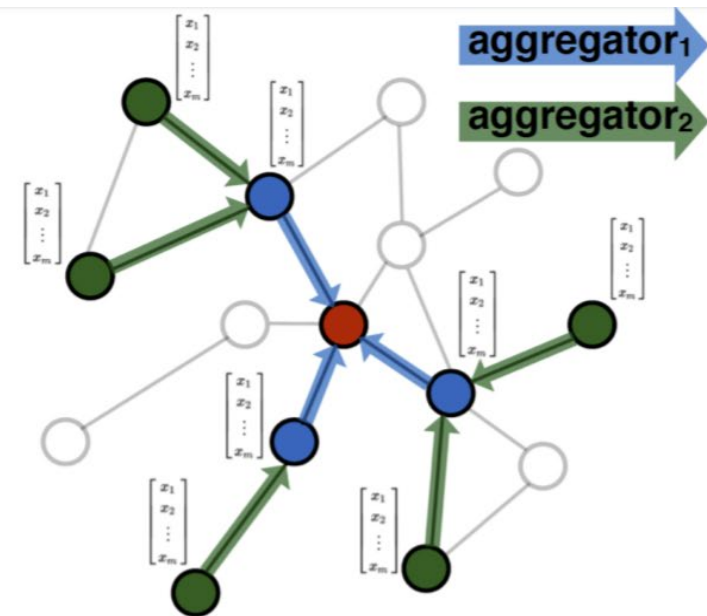
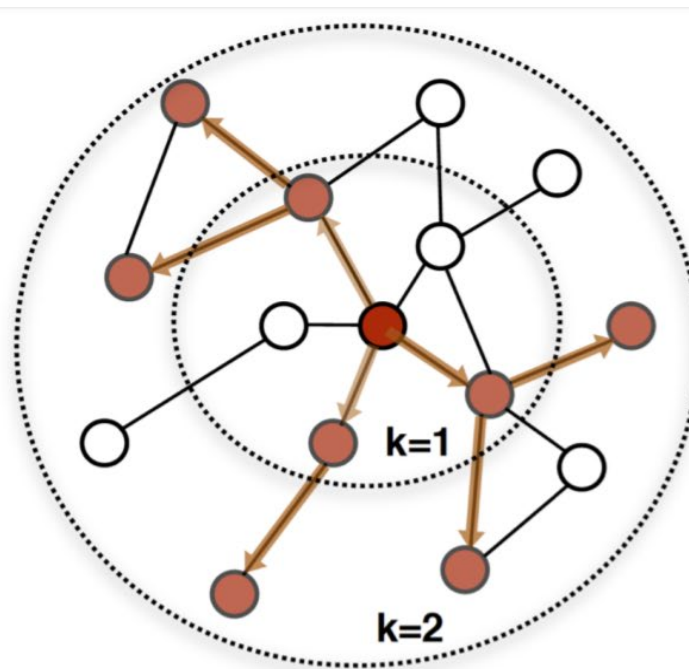
$$\mathbf{J} = \begin{bmatrix} \frac{\partial z_k^{(1)}}{\partial w_k^{(1)}} & \cdots & \frac{\partial z_k^{(1)}}{\partial w_k^{(u)}} \\ \vdots & & \vdots \\ \frac{\partial z_k^{(v)}}{\partial w_k^{(1)}} & \cdots & \frac{\partial z_k^{(v)}}{\partial w_k^{(u)}} \end{bmatrix}$$

Limitations of MPNNs

- **Long range dependencies:** Two distant nodes within the graph whose interaction is required.
- More MPNN layers required to cover distant interactions.
- ❖ Computationally expensive with growing receptive field.
- ❖ **Oversmoothing:** node features become homogeneous after multiple message passing iterations.
- ❖ **Oversquashing:** the MPNN output at a given node p is not sensitive to features of a node q that is k -hops away.

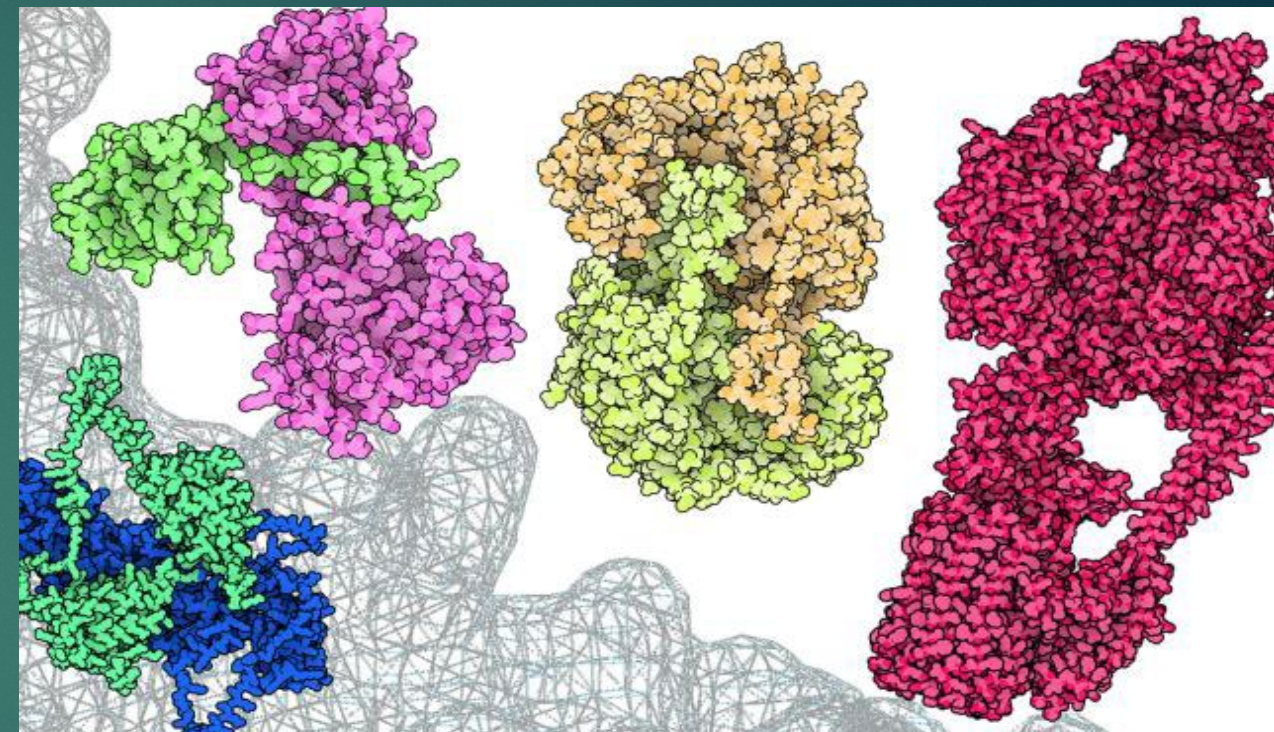


GraphSAGE



Supervised training to identify parameters: $W^{(k)}$, $Q^{(k)}$

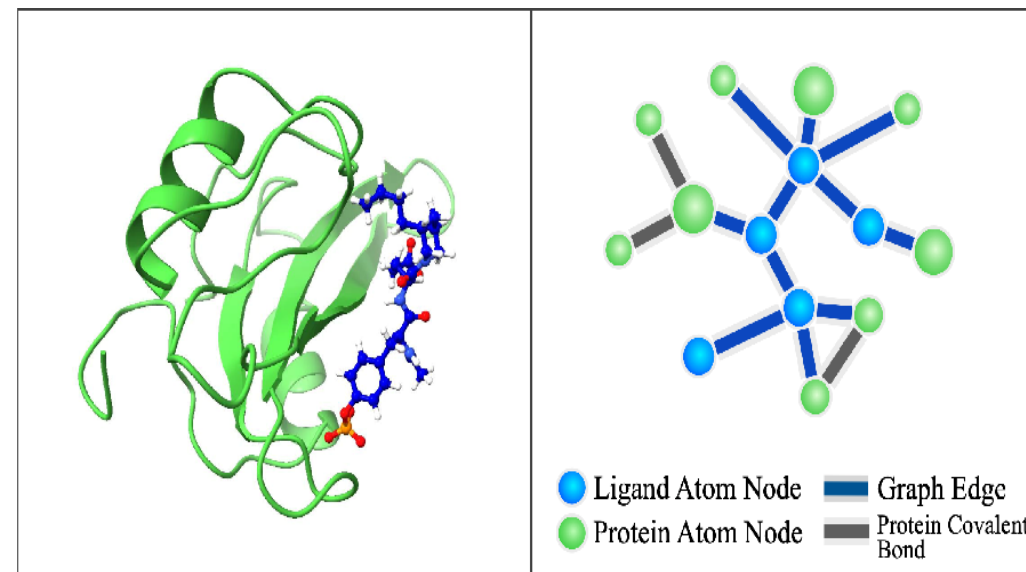
Applications in Life Sciences



GNNs for Protein Learning

Protein Representation through Message Passing

- **Message Passing Neural Nets (MPNNs):** Able to represent complex molecular surfaces and learn useful interactions among the atoms in those systems.
- The protein surface can be represented as a graph $G=(V,E)$.
 - Atoms/residues as nodes; Bonds as edges.
- The goal is to obtain a latent representation of the graph for residue-level or protein-level tasks.



(a) 3D Protein-Ligand Complex

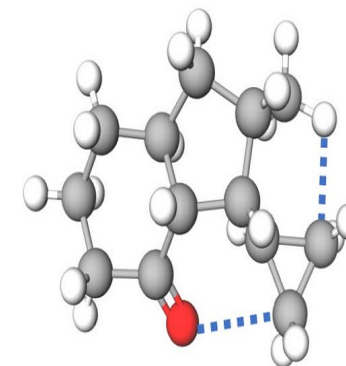
(b) Complex Interaction Graph

Limitations of MPNNs

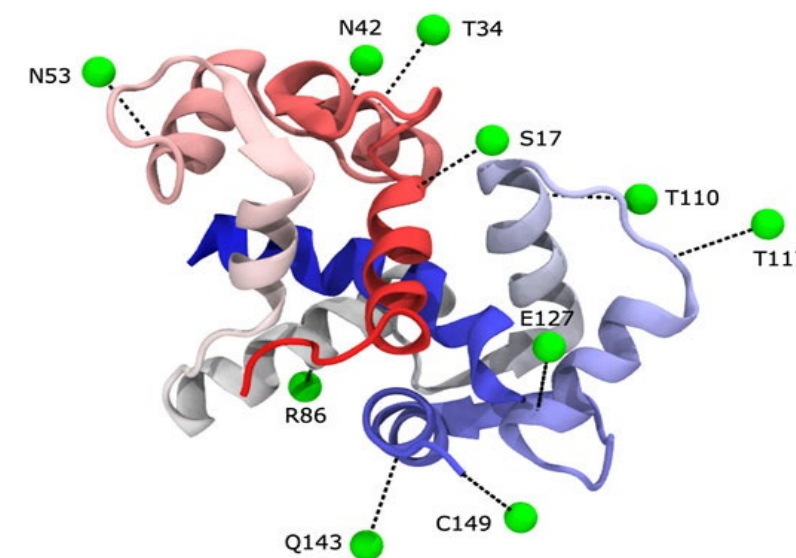
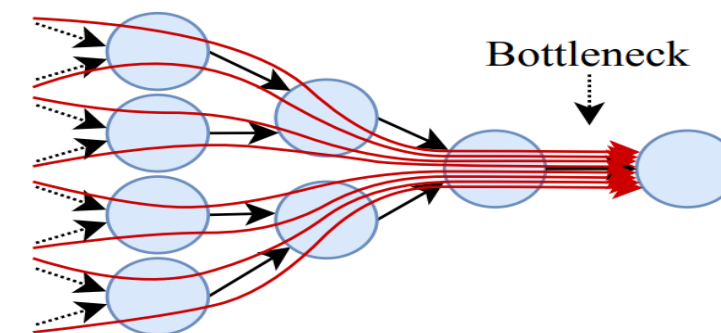
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❖ Long Range dependencies in Proteins

- Long-range interactions (LRI) can be crucial in protein structures.
- They can play a crucial role in **protein function regulation, stability and folding.**



A molecule with long range interactions.
The dotted lines show 3D atomic contact.

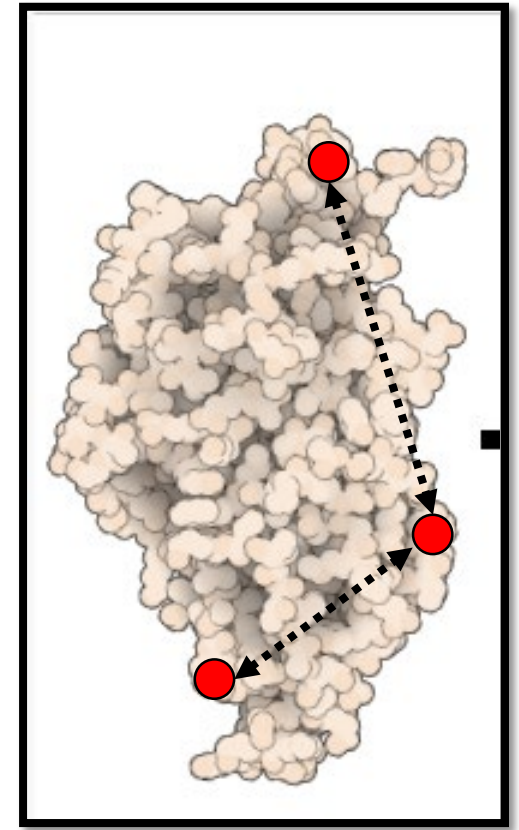


Objective

- Improve on current GNN methods that result in oversquashing and oversmoothing, especially for large-scale proteins.
- Develop a robust GNN model able to account for long-range dependencies for protein and residue-level tasks.

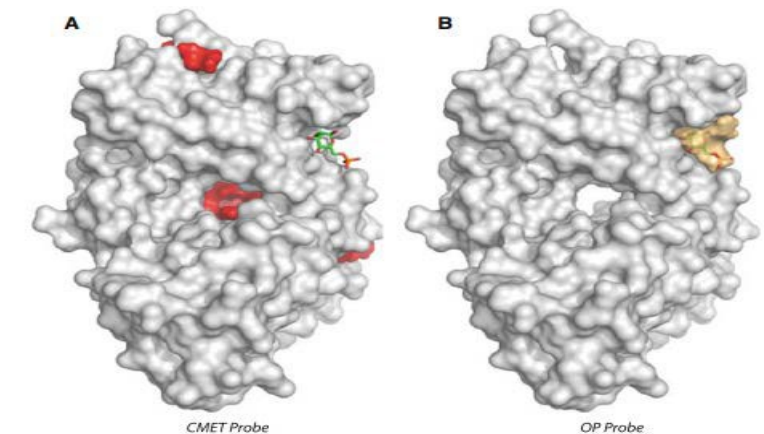
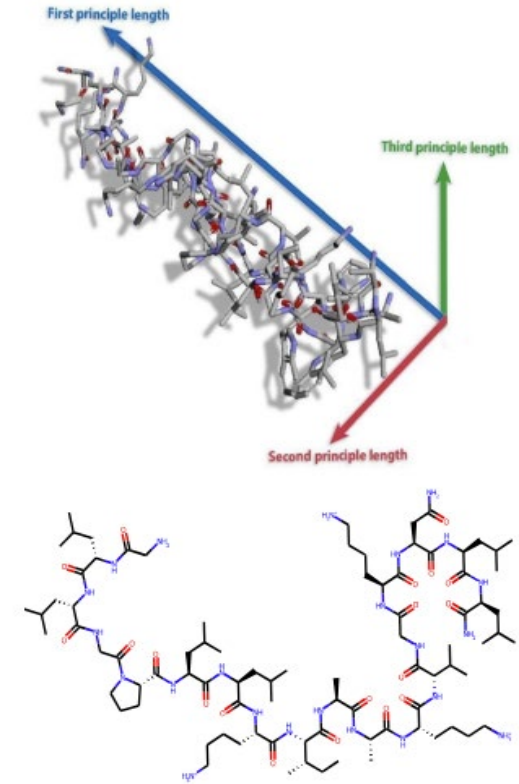
To proceed:

- Build on existing graph rewiring and bottleneck alleviation techniques to learn on proteins structures as graphs.



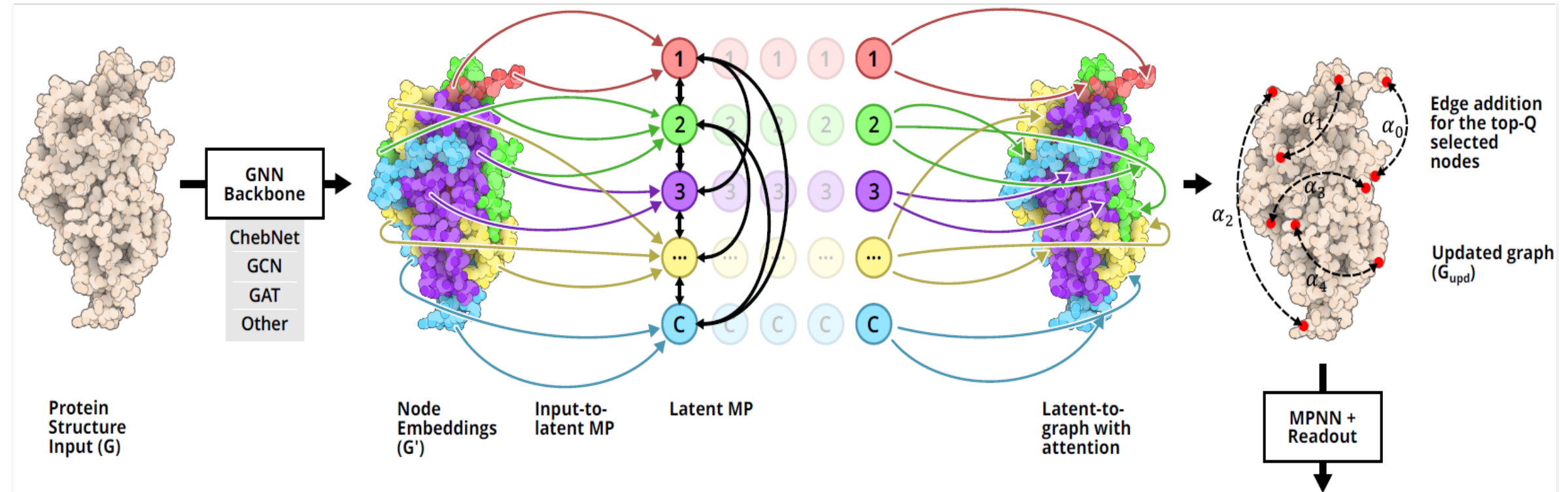
Datasets

- **Peptide functions:** Peptides are short chains of amino acids that play a crucial role in biological processes.
 - They exhibit distant dependencies through the long-range amino acid interactions that influence the chain.
- **Enzyme Classification:** It consists of predicting the type of reaction catalyzed by the given protein as given by the Enzyme Commission database.
- **Binding site identification:** on the PDBBind2020 dataset.



Method : Learning through rewiring via virtual nodes

- Distant nodes need to communicate for a more effective representation at the graph-level.
- We use attention as metric to quantify the importance of the interaction of 2 distant nodes using the latent nodes as a mediator.



Method: Learning on proteins through virtual nodes

Step 1: Aggregates information from the local neighborhoods of the protein

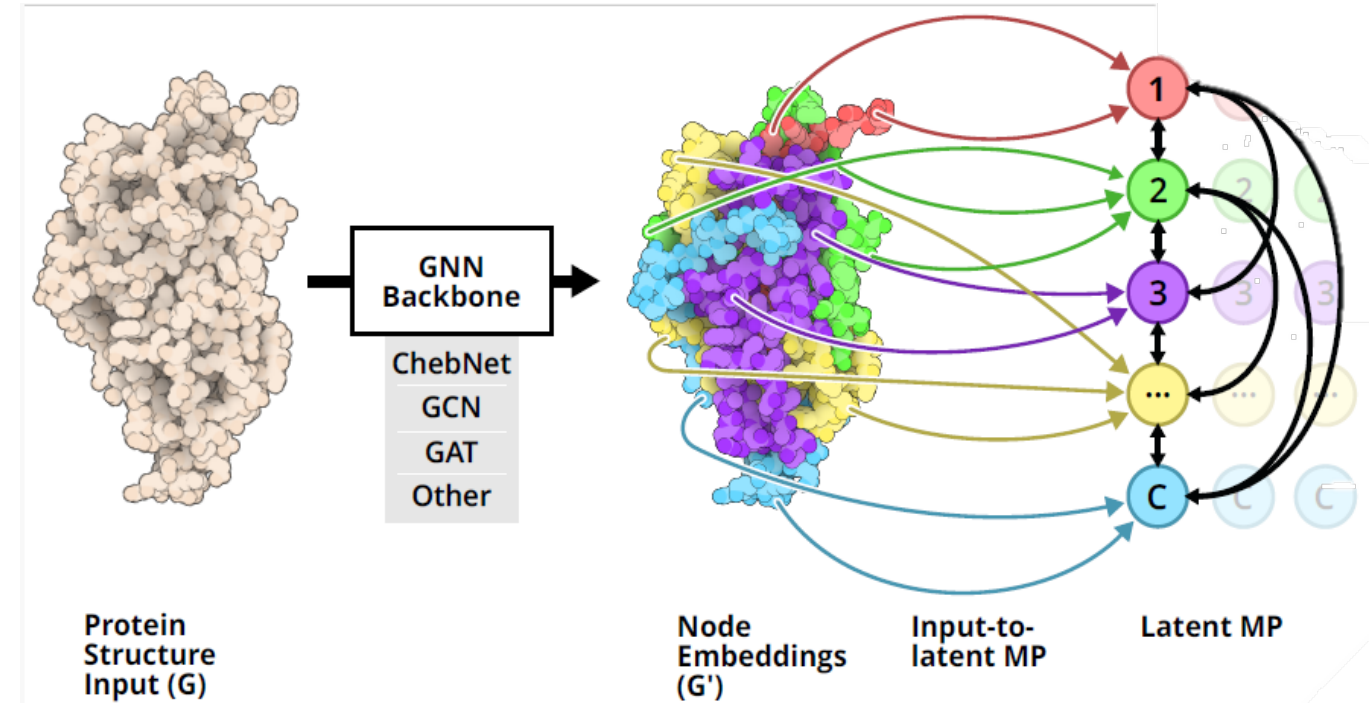
$$h_i^{(l+1)} = M_l(h_i^l, \sum_{j \in \mathcal{N}(i)} AGG_l(h_i^l, h_j^l))$$

Step 2: Maps different residues to virtual nodes randomly distributed in latent space.

$$z_q^{(l+1)} = M_l(z_q^l, \sum_{j \in \mathcal{N}(q)} AGG_l(z_q^l, h_j^l))$$

Step 3: Exchange message between virtual nodes, this serves as a solid global operator.

$$z_q^{(l+1)} = M_l(z_q^l, \sum_{j \in [1, c]} AGG_l(z_q^l, z_j^l))$$



Observations- Attention-based rewiring advantage

- On all the benchmark datasets, **GCN-Rewire performs better than both the GCN backbone** and the GCN model with multiple virtual nodes.
- On the Peptide dataset, it surpasses other rewiring frameworks such as FOSR and LASER.
- The method is successful in attending distant residues whose communication potentially determines the overall protein's function.

Model	Peptide-function	Peptide-structure
	Test AP $\uparrow$	Test MAE $\downarrow$
GCN	0.5930 ± 0.0023	0.3496 ± 0.0013
GPS Transformer	0.6535 ± 0.0041	0.2500 ± 0.0005
Expformer	0.6527 ± 0.0043	-
FOSR	0.4629 ± 0.0071	0.3078 ± 0.0026
LASER	0.6447 ± 0.0033	0.3151 ± 0.0006
Transformer + PE	0.6326 ± 0.0126	0.2529 ± 0.0016
GCN-Latent	0.6211 ± 0.0059	0.2723 ± 0.0040
GCN-Rewire	0.6670 ± 0.0024	0.2660 ± 0.0043

Table 1: Performance of our rewiring framework on the LRGB datasets against numerous baselines.

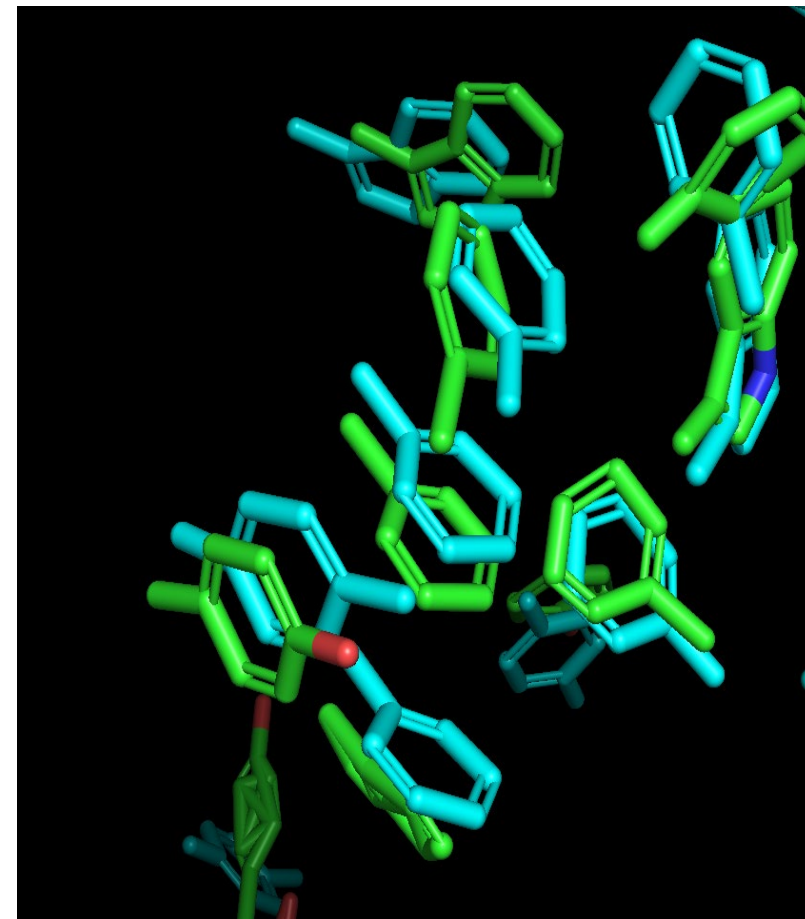
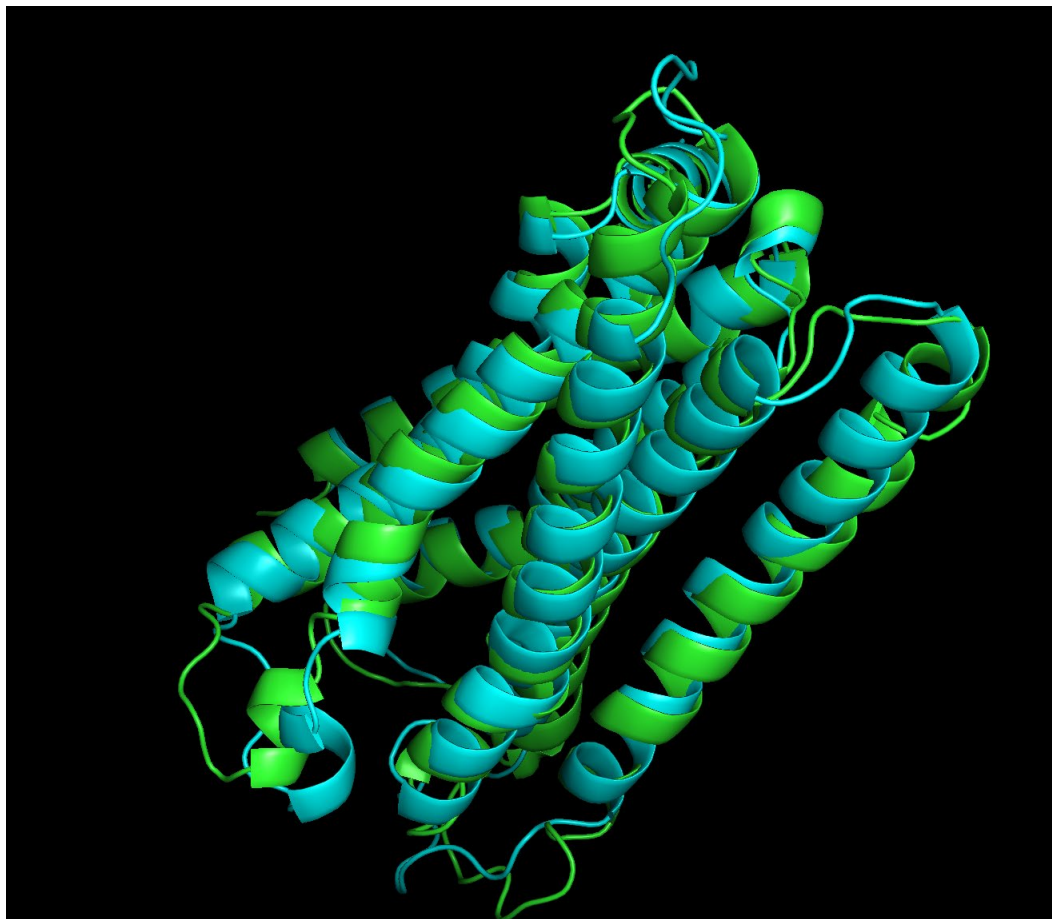
Model	EnzymesClass	PDBBind
Type	Graph-level	Node-level
	Test Acc $\uparrow$	Test AUROC $\uparrow$
GCN	73.33 ± 1.06	62.65 ± 0.13
GCN-Latent	74.22 ± 0.50	N/A
GCN-Rewire	75.80 ± 0.92	66.53 ± 0.34

Table 2: Performance of our rewiring framework on protein structure datasets.

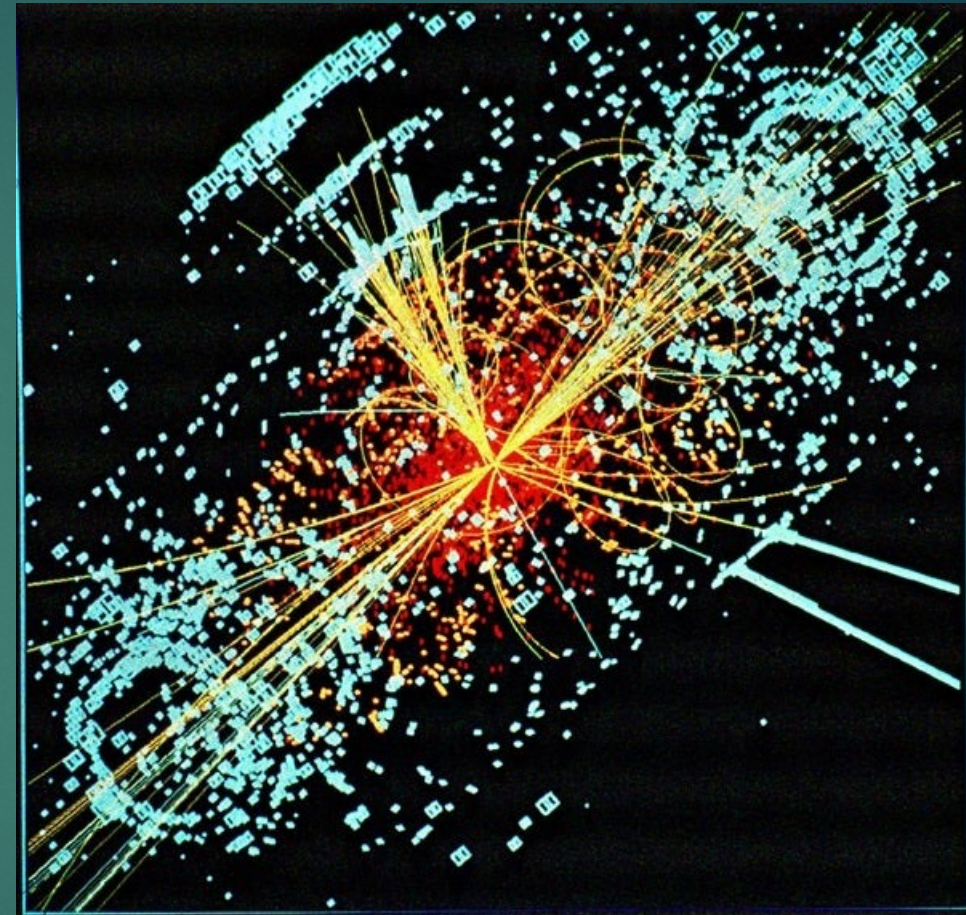
Future work

- We would like to have **further insight on the significance of the attention-based connections**, perhaps on more explainable datasets where the connections are known a priori.
- The connections are randomly initialized so far and the main goal is to cover the whole graph through virtual nodes. **A smarter strategy can be adopted.**

Other applications- Generative AI for Protein Design



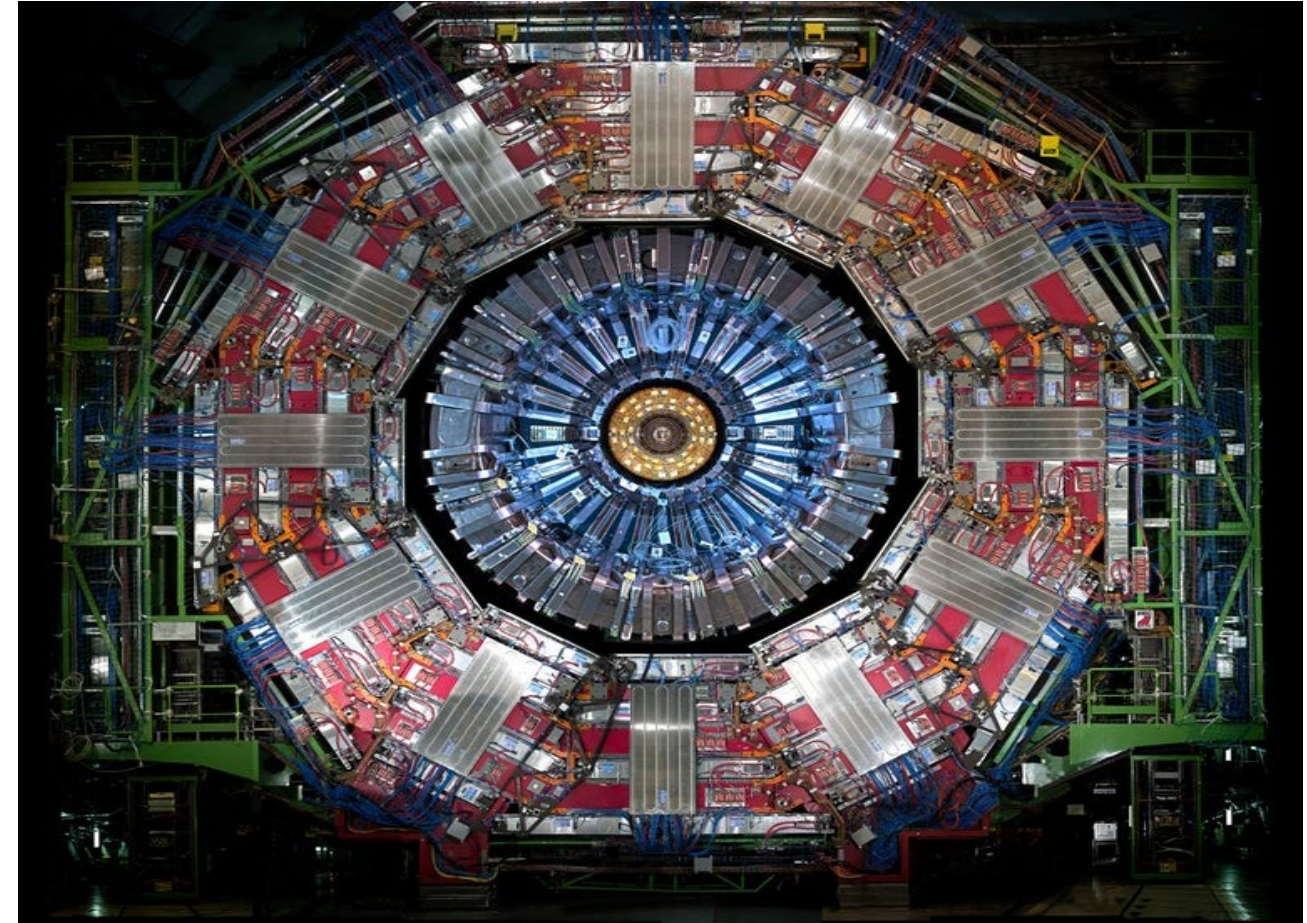
Applications in Life Sciences



GNNs for Particle Physics

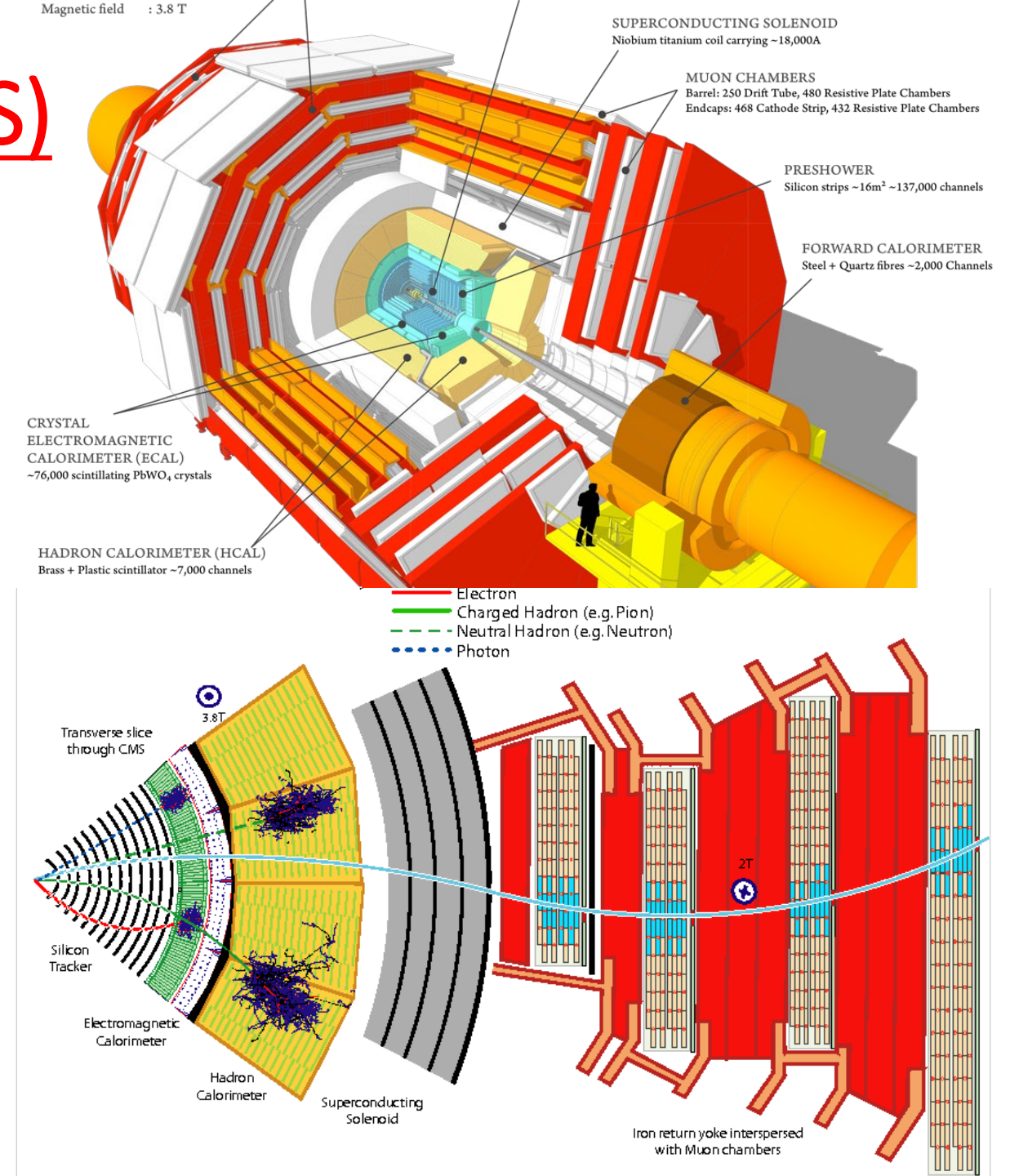
Compact Muon Solenoid (CMS)

- The Compact Muon Solenoid is a multi-purpose detector that is used to study particle properties following beam collisions.
- This machine consists of tightly packed materials that create a suitable operational environment for physics studies. detectors.
- The complete detector is **21 meters long, 15 meters wide and 15 meters high.**



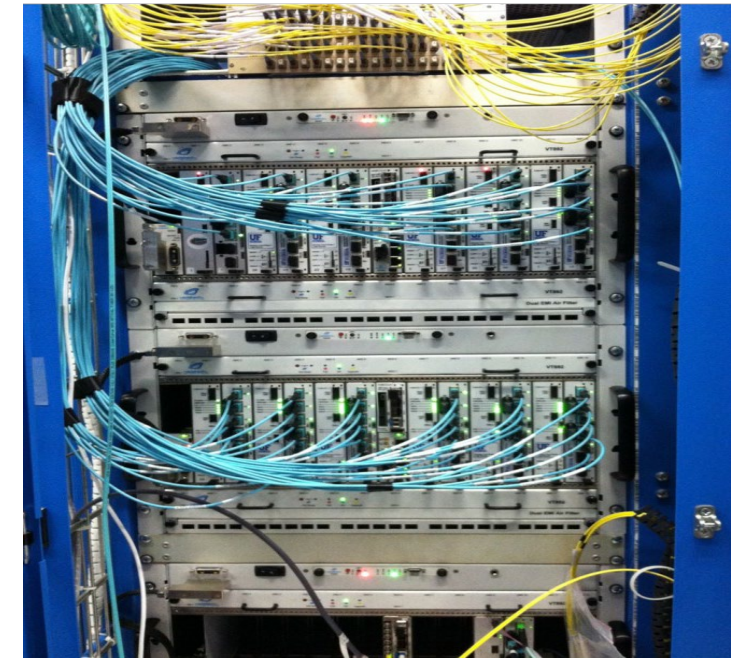
Compact Muon Solenoid (CMS)

- CMS has an “onion shape” enclosing overlapping sub-detector layers going outwards from the collision point.
- Crucial information about post-collision particles including their energy, momenta, and trajectories.
- Superconducting magnet provides a magnetic field of 4T, equal to 100, 000 times Earth’s magnetic field !
- Different particle types detected at different layers.

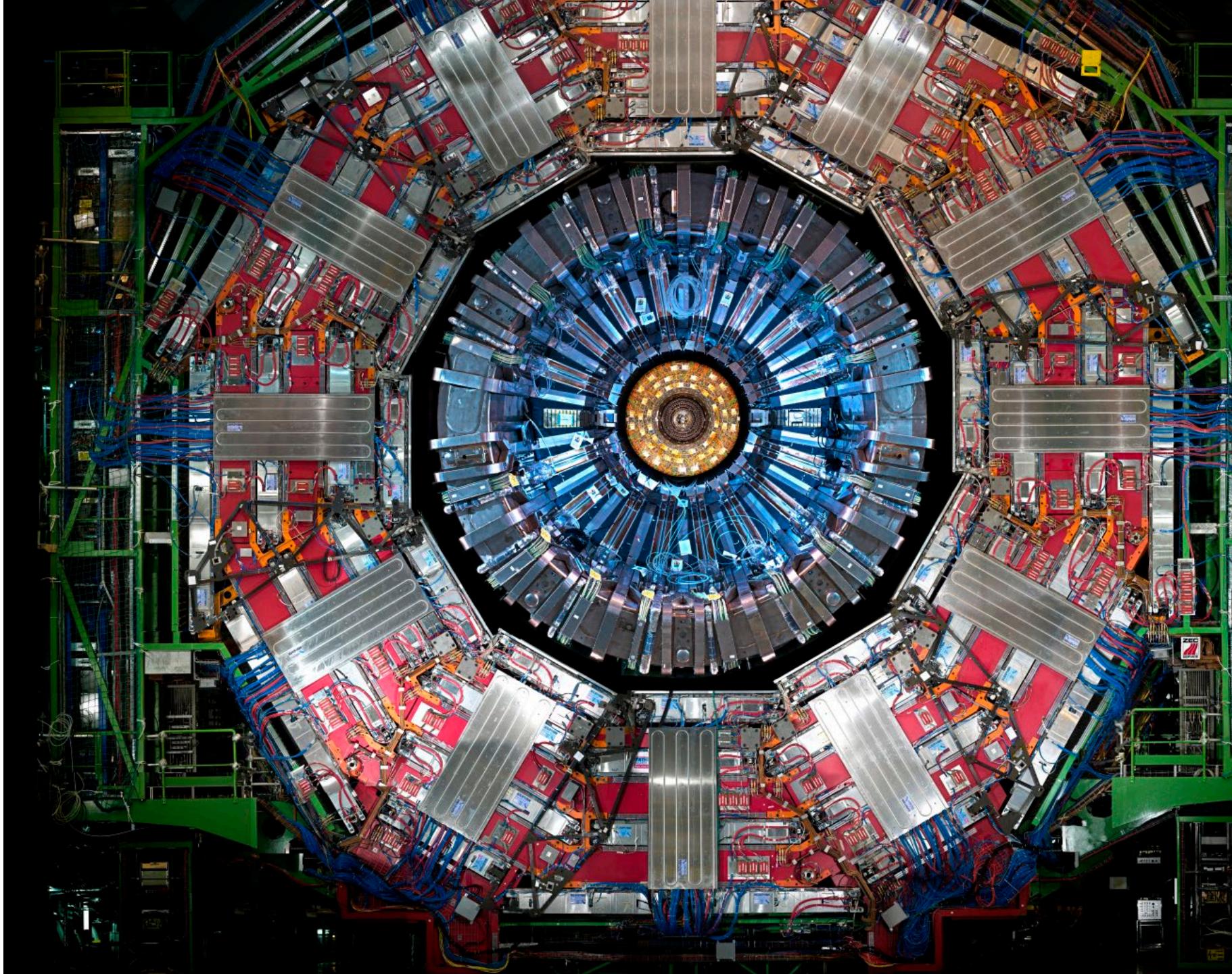


Deal with immense amounts of data

- LHC records events at a rate that is impossible for a human inspection to take place.
- Hardware components read billions of events and send 30 petabytes worth of data to the control center every year.
- Due to limited resources, SOTA algorithms are developed at CERN to filter events with a potential of uncovering new physics.
- For this purpose, CMS contains a trigger system that operates in several steps to accept or reject a post-collision event.

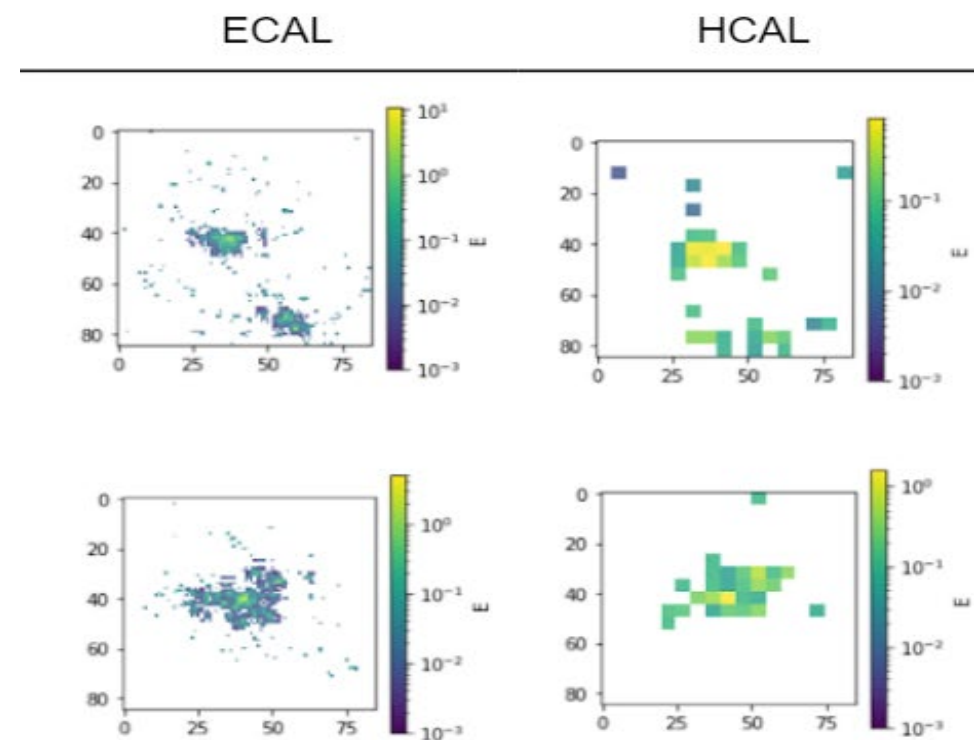


GNN Applications in Particle Physics



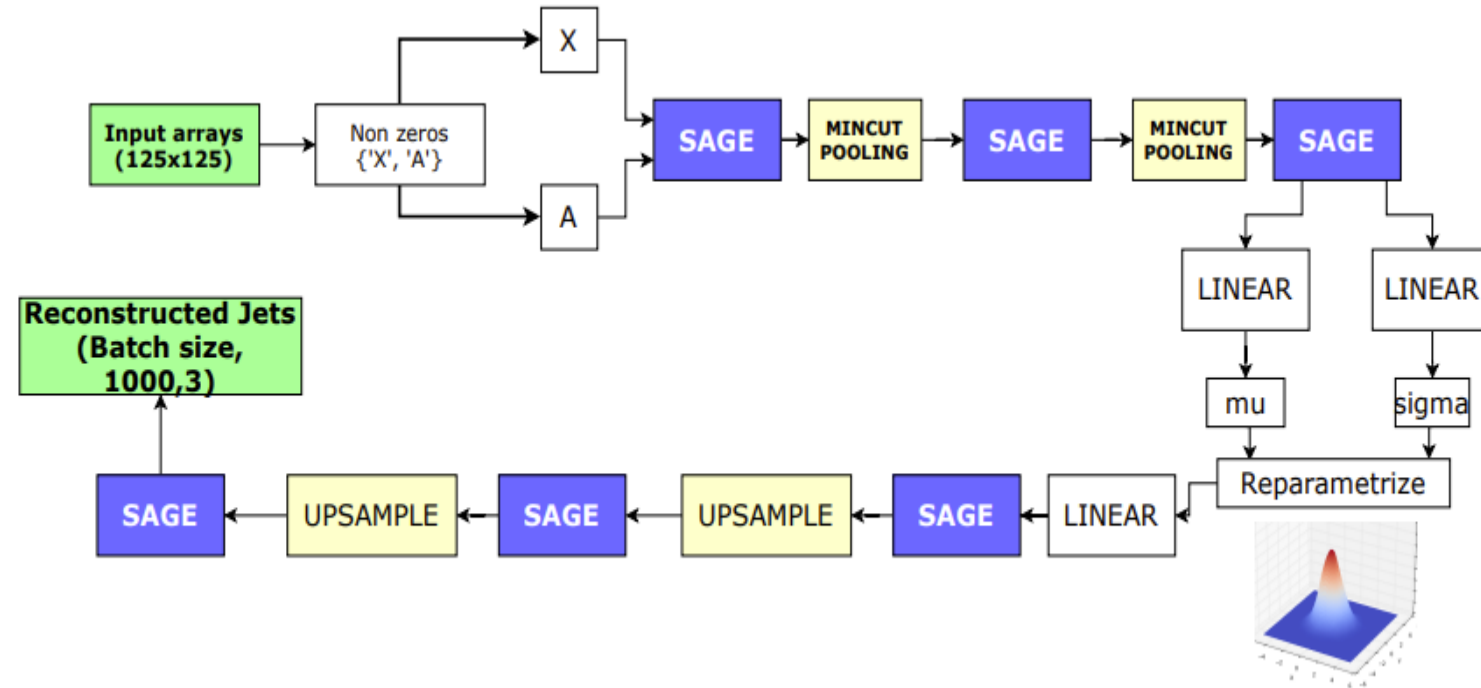
Graph VAE for Jet Reconstruction

- This work uses Open Data release public data from the LHC experiments. We use boosted top quark jets produced using Pythia 6, a program for generating particle collisions events.
- Data was transformed into image-based form, specifying the location and values of energy deposits in the calorimeter .
- Data consists of almost 30000 samples of $3 \times 125 \times 125$ arrays representing the mesh and the segmentation of 3 detector stages: **Tracker**, **ECAL** and **HCAL** subdetectors, respectively.

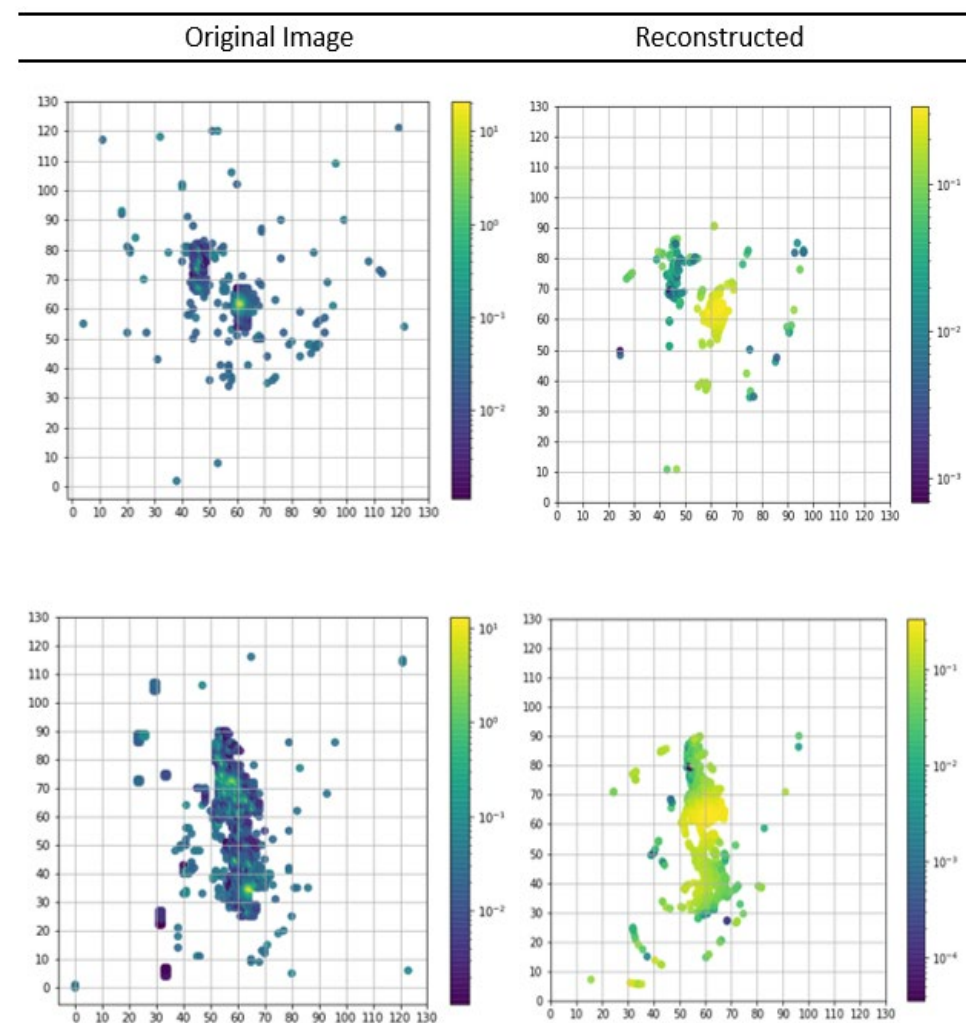
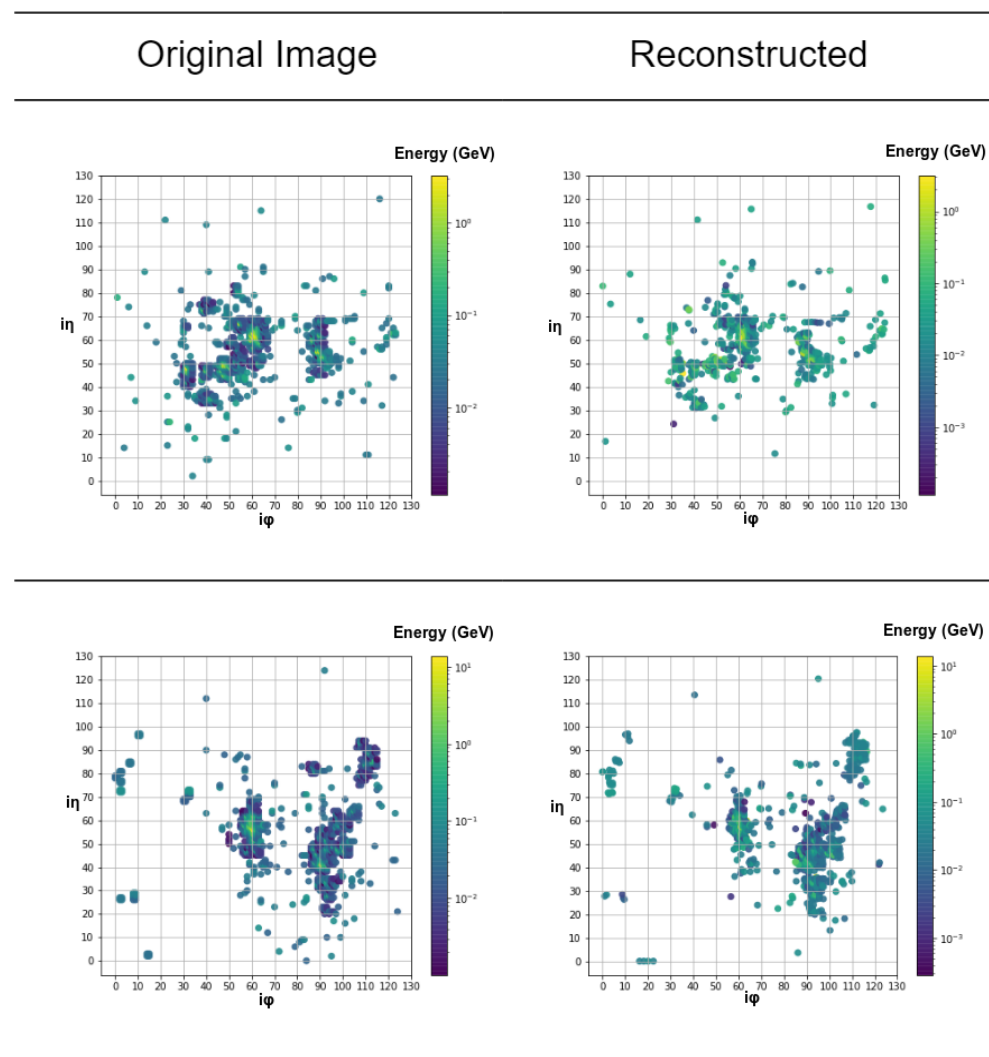


Graph VAE for Jet Reconstruction with MinCut Pooling

- Unlike in other fields, jets in particle collisions are not characterized by pre-defined topology.
- We connect each node to its k-nearest neighbors based on Euclidean distance.
- We aim to learn the properties of these jets as graphs in addition to their compressed representation.
- We develop a Graph VAE architecture whose encoder embeds the node features into latent space dimensions through Dense GraphSAGE layers.



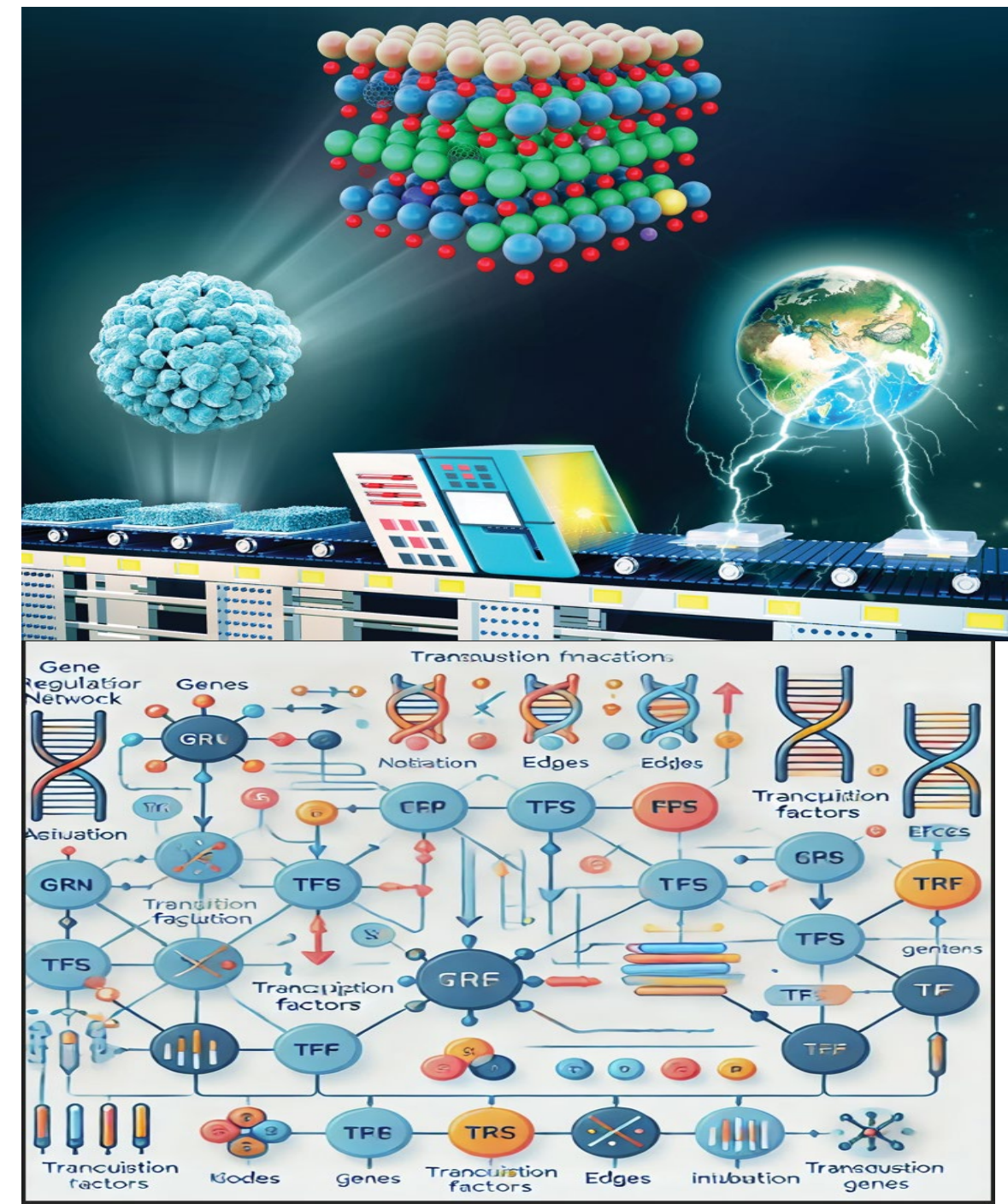
Mincut vs EdgePool



Model	Mincut Graph VAE	EdgePool GVAE	AE
Number of parameters	104 511	118 870	3 093 810

Future potential of long range-aware GNNs

- **Material science:** Crystal & Lattice Structures-
Predict material properties based on atomic arrangements with long-range dependencies.
- **Gene Regulatory Networks:** Identify distant genetic elements that influence gene expression.



Interested in rewiring graphs and other networks ?

LTS2 is the place to be !

Reach out via email: ali.hariri@epfl.ch

